

§11.2 - Series

Definition #1 (Infinite Series)

Let (a_n) be a sequence.

Then

$$\sum_{k=1}^{\infty} a_k = a_1 + a_2 + a_3 + \cdots$$

is an **infinite series**.

Definition #2 (Partial Sum)

Let (a_k) be a sequence.

Then

$$S_n = \sum_{k=1}^n a_k = a_1 + a_2 + \cdots + a_n$$

is the n th partial sum of the series.

Definition #3 (Convergence)

If

$$\lim_{n \rightarrow \infty} S_n = S \in \mathbf{R}$$

then the series $\sum a_n$ **converges**.

If (S_n) diverges then the series $\sum a_n$ **diverges**.

Remark

Here

1. $\sum a_n$ or $\sum a_k$: denotes the series.

2. $\sum_{n=1}^{\infty} a_n$: is the value of the series starting at $n = 1$.

3. $\sum_{n=2}^{\infty} a_n$: is the value of the series starting at $n = 2$.

Lemma #1

Let $r \in \mathbf{R}$.

Then

$$\lim_{n \rightarrow \infty} r^n = \begin{cases} 1 & r = 1 \\ 0 & r \in (-1, 1). \end{cases}$$

For any other r , the sequence diverges.

Definition #4 (Geometric Series)

Let $r \in \mathbf{R}$.

The geometric series

$$\begin{aligned}\sum_{k=0}^{\infty} r^k &= 1 + r + r^2 + \dots \\ &= \frac{1}{1 - r}\end{aligned}$$

exists only for $|r| < 1$.

Remark

Observe,

$$\sum_{k=0}^{\infty} 0^k = 1 + 0 + 0 + \dots$$

means we are defining $0^0 = 1$ in our summation.

Corollary #1

We have

$$\sum_{k=1}^{\infty} r^k = \frac{r}{1-r}$$

for $|r| < 1$.

Example #1

Find

$$\sum_{k=0}^{\infty} 2^{-k}.$$

Example #2

Find

$$\sum_{k=1}^{\infty} \frac{2}{(2k+1)(2k+3)}.$$

Theorem #1

If

$$\sum a_n$$

converges, then

$$a_n \rightarrow 0.$$

Corollary #2 (Basic Divergence Test)

Let (a_n) be a sequence.

If

$$\lim_{n \rightarrow \infty} a_n \neq 0 \quad \text{or} \quad \lim_{n \rightarrow \infty} a_n \quad (d.n.e.)$$

then

$$\sum a_n$$

diverges.

Example #3

Determine if

$$\sum n \sin \left(\frac{1}{n} \right)$$

converges.

Warning

There exists a sequence (a_n) such that

$$a_n \rightarrow 0 \quad \text{and} \quad \sum a_n \text{ diverges.}$$

Theorem #2

If $\sum a_n$ and $\sum b_n$ exist then

$$\sum_{k=1}^{\infty} (c \cdot a_k + b_k) = c \sum_{k=1}^{\infty} a_k + \sum_{k=1}^{\infty} b_k.$$

for any $c \in \mathbf{R}$.

Example #4

Express the numbers

$$0.1111111111\dots \quad \text{and} \quad 0.1212121212\dots$$

as ratios of integers.

Example #5

Find the values of x such that

$$\sum_{k=0}^{\infty} e^{kx}$$

converges.